

Total Precipitation Gauge

Overview

A python module for interacting with the Sutron Total Precipitation Gauge (TPG). The `tpgtochords` module is provided, which will send tpg readings and other system health metrics to a [CHORDS portal](#).

The TPG onboard logger is configured and controlled by commands sent via the RS-232 port.

The configuration is established by setting TPG variables. `tpg.py` sets a few of these at startup, to insure that the gauge measurement behavior is always consistent. It uses a default set of values, but these may be overridden by values in the `tpg.py` configuration:

TPG variable	tpg.py default	tpg.py config key
Station Name	n/a	station_name
Automeasure Interval	00:01:00	auto_time
Automeasure Time	00:00:00	auto_interval
Averaging Time	1.0	avg_time
Auto Output	0	

TPG readings are obtained by sending commands to the gauge, and parsing the returned output. The two key commands are **MEAS** (perform a measurement) and **TIME** (return the system time).

RS-232 Interaction

The TPG prompts with `>` after any command. The typical interaction will be to send a command, and read back lines until one containing just `>` is received.

Since the line with the prompt does not have a line terminator, the `readline()` will be issued with a timeout, so that the prompt line can be detected.

The text returned from the TPG is 'pretty formatted', and `tpg.py` has to dig through all of it to extract the desired values. This is not really too hard; the approach will be to create a list of tokens, search for matching tuples such as `("Precip", "0.0018")` or `("System", "time", "2017/12/28", "22:45:33")`, and capture the desired readings.

TPG Commands

These are the TPG commands used by `tpg.py`, and the associated output.

```
MEAS
Reading
    Precip 0.0018 in
    Precip in bucket 0.6648 in
    Field Cal Offset -0.6630057 ,    Precip Rate 0.0038 in/hour

Temp In Box 21.62 C

>
```

```
TIME
System time 2017/12/28 22:45:33
```

Set the time:

```
TIME 2017/12/28 22:45:33
```

```
BATT
Battery 11.8V
```

```
>
```

tpgtochords

This module matches a custom data system built around a Raspberry Pi Zero W, which performs the following:

- Once per minute:
 - Read the TPG measurement.
 - Read pressure, temperature and humidity from a BME280 sensor.
 - Read power information from an INA_219 sensor.
 - Transmit to CHORDS.

INA_219

- Uses this [INA_219 package](#).